IOTA 5201 · REINFORCEMENT LEARNING

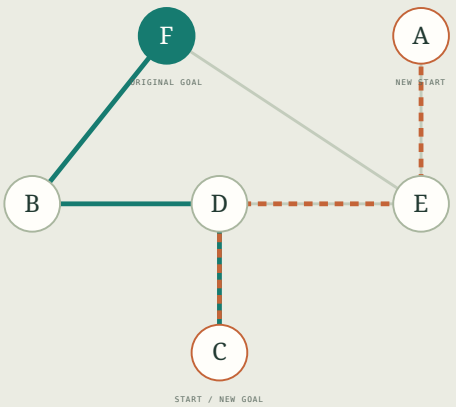
Q-learning, *room by room.*

Finding a good route is one question. Knowing whether the values are right is another.

A reproducible study of six rooms, two destinations, and the difference between a successful policy and a well-learned Q-table.

Chenglong Zhang
50068179 · 24 October 2026

THE SIX-ROOM WORLD 01 / ENVIRONMENT



Original C → F Modified A → C

A connectivity model redrawn from the assignment. Lines represent doorways, not physical distances.

06	02 + 04	120	20,000
rooms in the world	required + extension cases	seeded experiment runs	episodes per main run

01 / THE QUESTION

What does it mean to have learned the world?

Imagine a robot moving between rooms to reach its dock. It receives no map of future returns. Instead, it tries an action, observes a reward, and revises an estimate of how useful that action will be. This is the setting for tabular Q-learning.

The assignment asks us to compare an original start/target pair with a new one, **A → C**. I interpret the robot and dock in its figure as **C → F** for the original case. The prose does not explicitly identify that original pair, so this is a declared modelling assumption.

Our additional question is more diagnostic: **can the robot learn a shortest route while leaving other action values wrong?** We answer it using full-table error, coverage counts, repeated seeds, and an exploration ablation.

02 / THE METHOD

One transition.
One revised estimate.

A table entry Q(s, a) estimates the discounted return from taking action a in state s and acting optimally afterwards. Here an action names the adjacent room we enter.

We move the current estimate a fraction **α** towards a target: the immediate reward plus the discounted best legal next-action value. On entering the goal, there is no future action and the target is just the terminal reward.

The behaviour policy is **ε-greedy**: with probability ε it chooses uniformly among legal actions; otherwise it selects a maximum-Q action, breaking exact ties uniformly. The update uses a maximum even when exploration selects a different next action. This is the off-policy distinction described by Sutton and Barto [1].

THE UPDATE

$$Q(s, a) \leftarrow Q(s, a) + \alpha\,[y - Q(s, a)]$$

$y = r + \gamma \max_{b \in A(s')} Q(s', b)$

$y = r$ when s' is terminal

α controls the size of each revision.
γ makes earlier rewards more valuable.

03 / THE EXPERIMENT

A small world.
An explicit specification.

States and transitions

Each of A–F is a state. Doorways define deterministic, bidirectional edges. Unavailable moves are masked, never penalized as though they were valid actions. The terminal row has no outgoing actions.

ROOM	LEGAL DESTINATIONS BEFORE TERMINAL MASKING
A	E
B	D, F
C	D
D	B, C, E
E	A, D, F
F	B, E

This abstracts the room diagram into connectivity; it does not claim the rooms are equal-sized cells.

Rewards and stopping

Entering the target yields **+1**; every other move yields **0**. Episodes start at the declared fixed room and end at the target. A 100-step cap prevents unbounded sampling; a time-limit truncation still bootstraps because it is not a terminal goal.

With $\gamma = 0.9$ and zero intermediate rewards, maximizing discounted return favours the shortest route. We reset Q to zero for every configuration and seed; the modified case is trained from scratch.

SETTING	MAIN PROTOCOL
Q initialization	Zero on every valid pair
Learning rate α	0.1, constant
Discount γ	0.9
Exploration ε	0.5, constant; ablation 0
Budget	20,000 episodes × 20 seeds
Episode cap	100 transitions
Random generator	NumPy default_rng / PCG64
Run environment	Python 3.13.12 · NumPy 2.5.3

> **Why this budget?** A retained pilot used ε = 0.2 and 3,000 episodes. All 20 runs in each required case found a shortest route, but worst full-table errors were 0.491 (C → F) and 0.093 (A → C). We increased ε to 0.5 and the budget to 20,000 episodes to improve coverage. Both settings changed, so the pilot is a motivation and coverage warning, not an isolated estimate of either setting's effect. Its complete outputs are retained in `experiments/pilot_3000/`.

Six cases, one shared protocol

CASE	START → TARGET	PURPOSE
Original	C → F	required; original pair inferred from the figure
Modified	A → C	required A-to-C configuration
Start control	A → F	extension: hold target F fixed
B to F	B → F	extension: factorial comparison

CASE	START → TARGET	PURPOSE
B to C	B → C	extension: factorial comparison
No forced exploration	C → F	extension: diagnose false stabilization

Seeds 0–19 are separate runs. The same seed list is used across configurations, but trajectories consume different numbers of random draws: matching seed labels does not make transitions identical. Seed 0 is the preselected illustrative run throughout. All 120 main runs use 20,000 episodes and the same α, γ, rewards, and cap. Only the ablation changes ε.

04 / REQUIRED RESULTS

The route is short. The evidence goes further.

VALUE ACCURACY

Full-table error

$E_e = \max_{(s,a)\ \text{valid}} |Q_e(s,a) - Q^*(s,a)|$, measured after episode e. It includes actions away from the learned route.

OBSERVED STABILITY

Episode change

$\Delta_e = \max_{(s,a)\ \text{valid}} |Q_e(s,a) - Q_{e-1}(s,a)|$. Small changes can reflect either accurate values or infrequent updates.

A CHECKABLE REFERENCE

Exact Q*

$Q^*(s,a) = 0.9^{d(a,g)}$, where d is the shortest distance from the destination of action a to goal g. Goal entry has d = 0, so Q* = 1.

For this deterministic model, a breadth-first search supplies the exact reference independently of training. A separate value-iteration check agrees for every goal. The learner never uses this oracle for action selection or updates. We call a run **empirically accurate at tolerance 10^{-3}** when E remains below that threshold for at least 100 episodes through the end of the run. This criterion is retrospective; we still use the full fixed budget.

REQUIRED CONFIGURATION 01

Original · C → F

`C → D → B → F`

All **20/20 seeds** met the full-table accuracy criterion. Median qualifying episode: **1,119**, with a 10th–90th percentile range of 893.3–1,734.7. Every final greedy policy reached the goal in **3 steps**, matching the shortest path. Maximum final error across seeds: 1.78e-15; the least-visited valid pair across all seeds still received 2,366 updates.

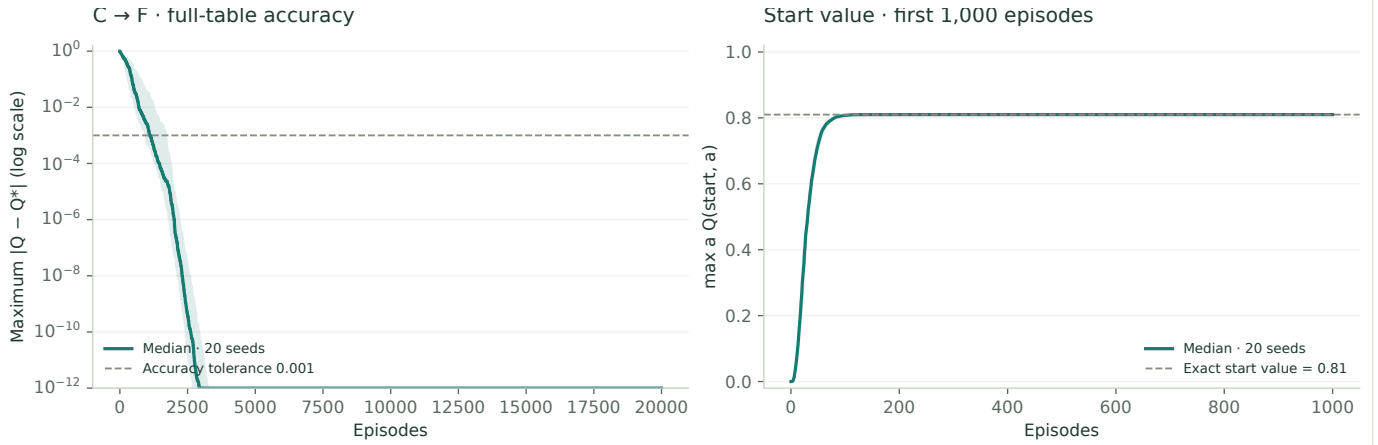


Figure 1. C → F over 20 seeds: full-table error (first panel) and start value during the first 1,000 episodes (second panel). Lines are medians; bands span the 10th–90th seed percentiles. Numerical source: results/original/aggregate.csv. Full-size figure.

Initial Q-table · episode 0

C → F, seed 0 · initial q-table · episode 0

S / A	A	B	C	D	E	F
A	–	–	–	–	0.000000	–
B	–	–	–	0.000000	–	0.000000
C	–	–	–	0.000000	–	–
D	–	0.000000	0.000000	–	0.000000	–
E	0.000000	–	–	0.000000	–	0.000000
F	–	–	–	–	–	–

Final Q-table · episode 20,000

C → F, seed 0 · final q-table · episode 20,000

S / A	A	B	C	D	E	F
A	–	–	–	–	0.900000	–
B	–	–	–	0.810000	–	1.000000
C	–	–	–	0.810000	–	–
D	–	0.900000	0.729000	–	0.900000	–
E	0.810000	–	–	0.810000	–	1.000000
F	–	–	–	–	–	–

Rows are current rooms; columns are destination actions. “—” means unavailable, including every outgoing action from target F. Initial CSV · Final CSV · Visit counts · Plot data

---

REQUIRED CONFIGURATION 02

Modified · A → C

A → E → D → C

All **20/20 seeds** met the full-table accuracy criterion. Median qualifying episode: **767.5**, with a 10th–90th percentile range of 627.1–901.1. Every final greedy policy reached the goal in **3 steps**, matching the shortest path. Maximum final error across seeds: 1.78e-15; the least-visited valid pair across all seeds still received 2,116 updates.

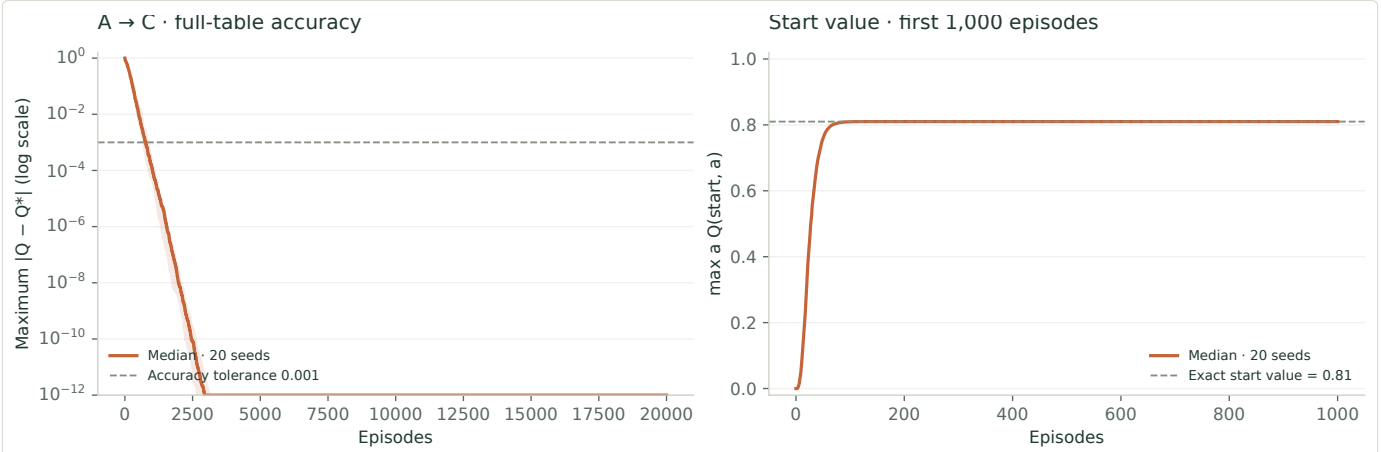


Figure 2. A → C over 20 seeds: full-table error (first panel) and start value during the first 1,000 episodes (second panel). Lines are medians; bands span the 10th–90th seed percentiles. Numerical source: results/modified/aggregate.csv. Full-size figure.

Initial Q-table · episode 0

A → C, seed 0 · initial q-table · episode 0

S / A	A	B	C	D	E	F
A	–	–	–	–	0.000000	–
B	–	–	–	0.000000	–	0.000000
C	–	–	–	–	–	–
D	–	0.000000	0.000000	–	0.000000	–
E	0.000000	–	–	0.000000	–	0.000000
F	–	0.000000	–	–	0.000000	–

Final Q-table · episode 20,000

A → C, seed 0 · final q-table · episode 20,000

S / A	A	B	C	D	E	F
A	–	–	–	–	0.810000	–
B	–	–	–	0.900000	–	0.729000
C	–	–	–	–	–	–
D	–	0.810000	1.000000	–	0.810000	–
E	0.729000	–	–	0.900000	–	0.729000
F	–	0.810000	–	–	0.810000	–

Rows are current rooms; columns are destination actions. “—” means unavailable, including every outgoing action from target C. Initial CSV · Final CSV · Visit counts · Plot data

How to read the plots. Left: median error across 20 seeds, with a shaded 10th–90th percentile band and the 10^{-3} tolerance. The logarithmic error axis is floored at 10^{-12} only for display; exact values remain in the CSV. Right: the maximum Q-value at the configured start, with the same across-seed band and a dashed exact reference. Episode 0 is initialization. Bands describe seed variation, not confidence intervals. Tables show seed 0 to six decimals; the local CSVs retain numerical precision.

A policy can tie without being wrong

For target F, D → B → F and D → E → F are equally short. We break evaluation ties alphabetically when values are exactly equal; small numerical differences may choose a different route. During training, ties are random. The criterion is success and shortest-path length, not a single prescribed sequence.

05 / CONTROLLED COMPARISONS

Change the start. Then change the goal.

The required comparison changes two variables together. To separate them, we add A → F, B → F, and B → C. Together with A → C, these form a 2 × 2 design: starts A/B and targets F/C. Using B avoids the degenerate C → C episode, which would terminate before taking any action.

With a fixed goal, the optimal Q-table depends on transitions, rewards, and discounting; it does not depend on which start we train from. Start choice still affects which actions receive updates and how quickly the full table becomes accurate. Changing the goal changes terminal boundaries and the values themselves.

START → TARGET	ACCURACY EPISODE: MEDIAN [P10, P90]	MEDIAN TRANSITIONS	FEWEST VISITS, ANY SEED	SHORTEST ROUTE
C → F	1,119 [893.3, 1,734.7]	87,309	2366	3 steps
A → F	4,769 [3,055.7, 6,101.4]	61,598	435	2 steps
B → F	6,286.5 [1,776.5, 6,761.1]	36,693	483	1 steps
A → C	767.5 [627.1, 901.1]	107,937	2116	3 steps
B → C	1,103.5 [926, 1,405.8]	82,640	1789	2 steps

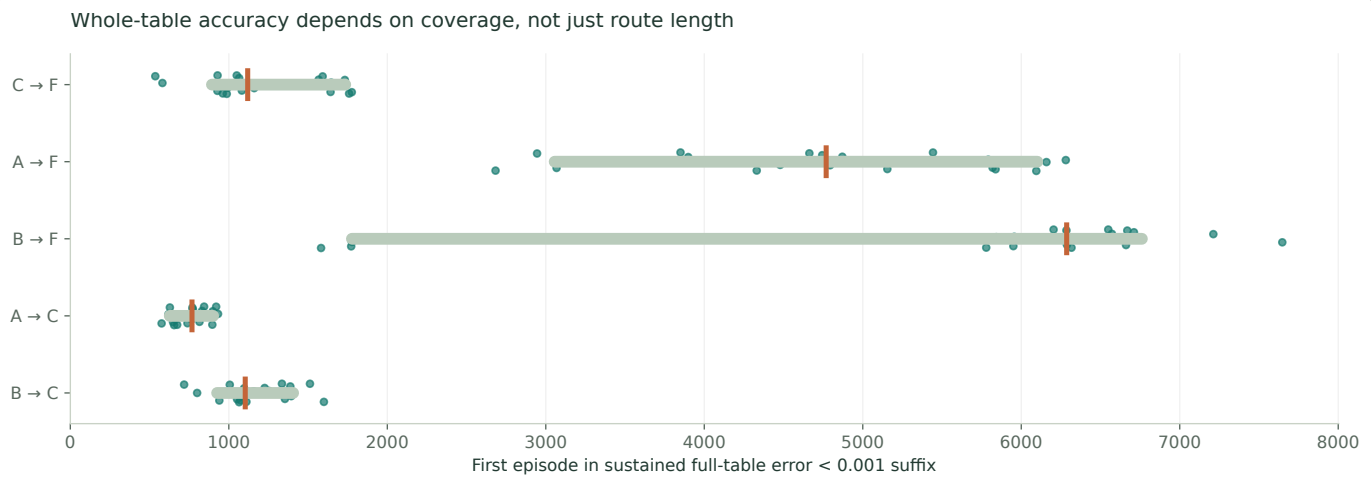


Figure 3. Each dot is one seed's first episode in a below-10^{-3} error suffix that persists to the end for at least 100 episodes. Thick marks show medians. Cases share hyperparameters and episode budgets. This diagnoses sample coverage, not general task difficulty.

A shorter route can leave a slower table.

Holding the goal at F, changing the start from C to A reduces the shortest route from three steps to two. Yet median full-table accuracy moves from 1,119 to 4,769 episodes. The descriptive seed-matched difference is 3,573.5 [2,061.3, 4,774.1] episodes (median [P10, P90]). Starting nearer the goal gives fewer opportunities to sample distant branches in each episode. This interpretation is consistent with the saved visit counts; it is not a universal law about start distance.

The target changes the learning problem.

Holding the start at B, switching target F to C increases the shortest route from one step to two, yet median full-table accuracy improves from 6,286.5 to 1,103.5 episodes. Goal F terminates on entering a well-connected room, while reaching leaf C requires passing through D. Coverage and termination structure matter beyond path length. Final reference tables for the same goal agree regardless of start, while changing the goal changes valid terminal actions and values.

Episode budgets are matched, but exploratory trajectories have different lengths. The table also reports median transition counts so that a comparison in episodes is not mistaken for equal computational or interaction cost. Seed-level results are available in summary.csv. This is a small descriptive study, with no significance claim or claim that 20 seeds characterize every possible random trajectory.

06 / OUR EXTENSION

A quiet table can still be wrong.

We repeat C → F with **ε = 0**. Greedy ties are still broken randomly, so this does not mean the run has no initial randomness. Once a route has positive estimates, however, lower-valued alternatives may receive no further attention.

In these saved runs, **20/20 final greedy policies take a shortest route**. Nevertheless, final full-table errors range from **0.900 to 1.000**, and 0–7 valid action pairs per run were never visited. No run meets the full-table tolerance. Episode changes remain below 0.001 from a median episode of **77** onwards for at least 100 episodes through the run end.

This is why success rate alone is insufficient evidence that the entire Q-table has converged. The robot can execute one good route while being poorly informed about actions it rarely takes.

NO FORCED EXPLORATION · 20 SEEDS

20/20

shortest-route success

versus incomplete value learning

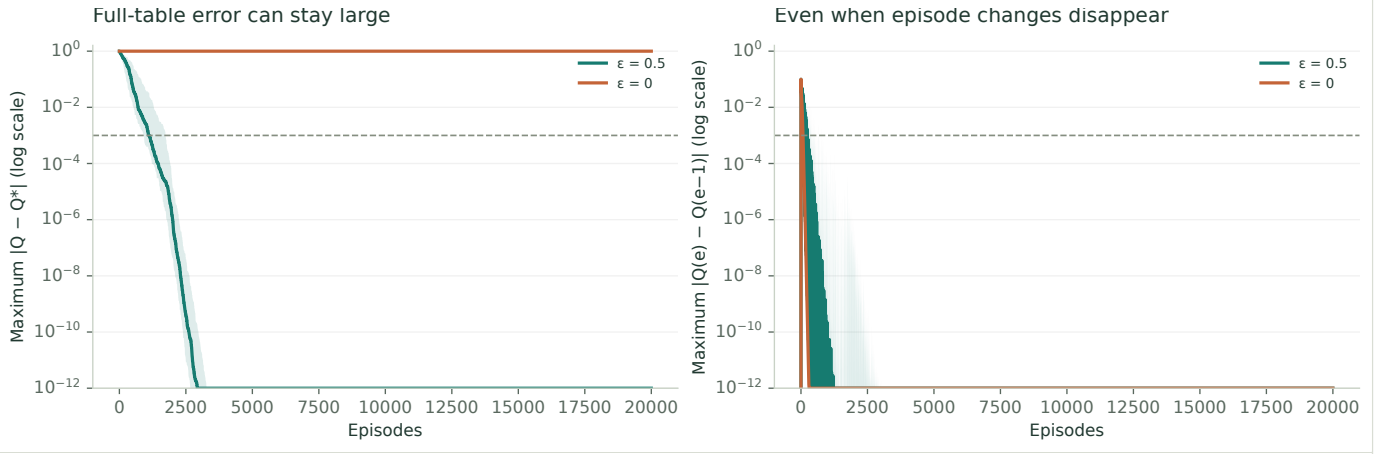


Figure 4. C → F with ε = 0.5 versus ε = 0, with other settings held fixed. Both panels use median and 10th–90th percentile bands across 20 seeds; left is error against Q*, right is the change in the Q-table during each episode. Values below 10^{-12} are floored for the logarithmic display.

Stabilization is an observation, not a theorem

We also save the Bellman residual, max |T*Q − Q|, and every state-action visit count. The residual checks consistency with the known model; the counts reveal coverage. In the main required runs, the largest final residual across all 40 runs is 5.55e-16, and every valid pair has at least 2,116 visits. These checks strengthen a finite-run accuracy claim.

The classical convergence result requires repeated sampling and suitable decreasing step sizes [2]. Our constant α = 0.1 and finite budget do not establish those asymptotic conditions. Deterministic transitions make a precise reference possible here; noisy rewards, larger state spaces, alternative graph interpretations, and other hyperparameters could change the picture.

07 / TAKEAWAYS

Three lessons to carry forward.

01

The goal defines value.

Moving the target changes rewards and the terminal boundary. A new goal generally requires a new value function.

02

The start shapes experience.

Under the same goal, different starts share Q* but expose the learner to different trajectories and coverage bottlenecks.

03

Check more than success.

A good route, a stable table, and an accurate table answer different questions. Compare all three before claiming convergence.

08 / REPRODUCE & VERIFY

Every figure has a trail.

The project includes the learner, explicit configurations, all 120 main-run traces, pilot evidence, seed-level metadata, Q-tables, reference values, visit counts, and plotting source. Essential text, tables, and figures are local and require no training run or external service to read.

Personal repository · clzJY/iota-qlearning ↗
Download submission ZIP ↗ · Read the PDF
Experiment configuration · All-run summary

The online release links require a connection. The submitted ZIP contains the complete offline report and the files needed to reproduce the evidence.

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m unittest discover -s code -p 'test_*.py'
python code/run_experiments.py
python scripts/build_report.py
python scripts/verify_project.py --replay
python scripts/package_submission.py
```

Exact main commands; see the README for individual configurations, clean output directories, package versions, optional PDF generation, and local serving.

References and attribution

1. Sutton, R. S. & Barto, A. G. *Reinforcement Learning: An Introduction*, 2nd ed., Chapter 6.5, Q-learning. Author's book page ↗
2. Watkins, C. J. C. H. & Dayan, P. (1992). Q-learning. *Machine Learning* 8, 279–292. Author's paper page ↗
3. Yu, J. (Fall 2026). *Mid-Term Assignment: Implementing Q-Learning*, IOTA 5201. Course-provided assignment brief. The room connections and original C → F interpretation are derived from its Figure 1. Our diagrams are newly drawn; the robot photographs are not reused.
4. GitHub Docs. Configuring a publishing source for GitHub Pages ↗. Hosting workflow only.

AI assistance and responsibility

OpenAI Codex helped interpret the brief, draft the Python implementation and checks, run experiments, build figures and the HTML report, and prepare reproducibility tools. Verification performed during development included numerical unit checks, an independent value-iteration comparison, saved-artifact checks, seed replay, and browser checks. All numerical claims above are generated from saved Python runs, not illustrative AI output.

This first version was produced with AI assistance. These checks do not establish the student's understanding. Chenglong Zhang remains responsible for reviewing the model, code, evidence, explanations, and AI-use statement before submission.

Chenglong Zhang · 50068179